

Training Course

Module 09: Configure Azure Monitor



Agenda



Lesson 01: Configure Azure Monitor



Lesson 02: Configure Azure Alerts



Lesson 03: Configure Log Analytics



Lesson 04: Configure Network Watcher

Lesson 01: Configure Azure Monitor



Describe Azure Monitor Key Capabilities



Monitor & Visualize Metrics

Metrics are numerical values available from Azure Resources helping you understand the health, operation & performance of your systems.

[Explore Metrics](#)



Query & Analyze Logs

Logs are activity logs, diagnostic logs and telemetry from monitoring solutions; Analytics queries help with troubleshooting & visualizations.

[Search Logs](#)



Setup Alert & Actions

Alerts notify you of critical conditions and potentially take corrective automated actions based on triggers from metrics or logs.

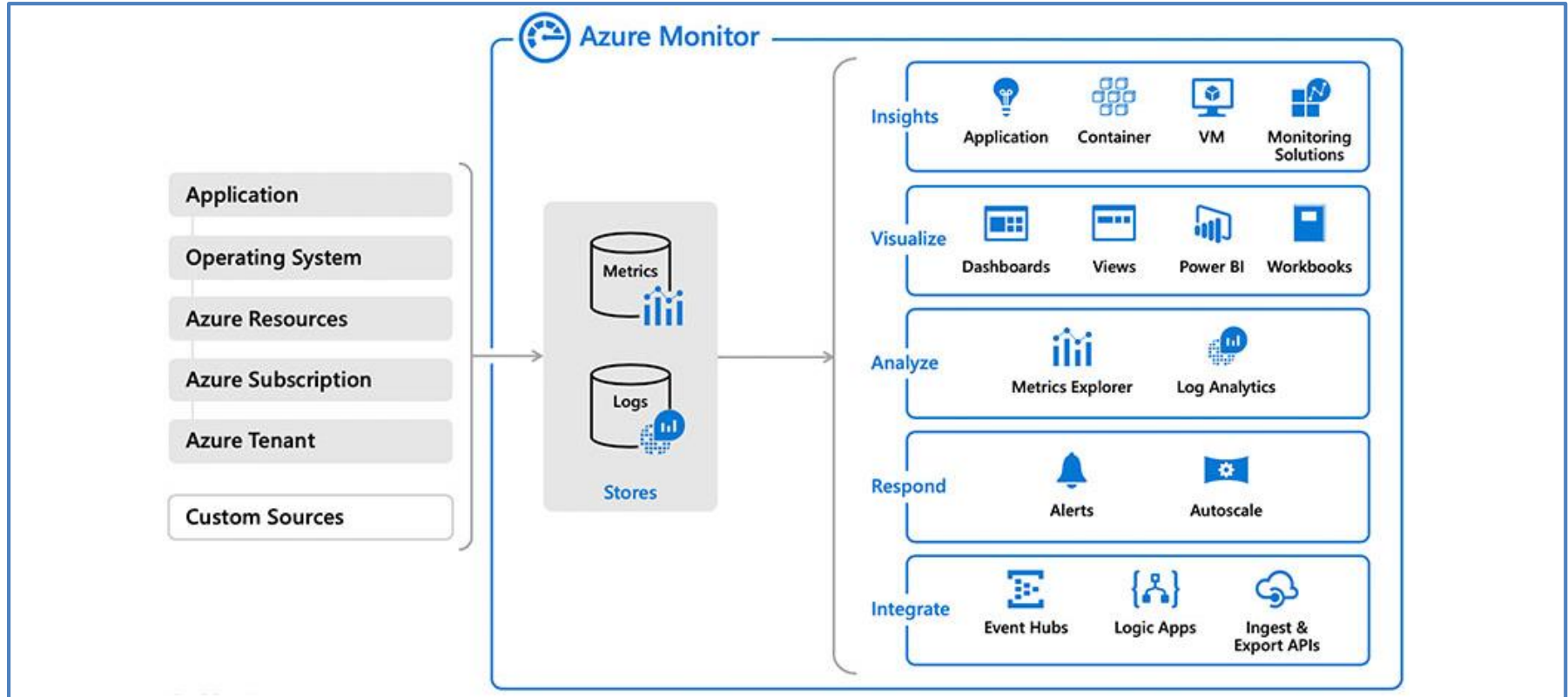
[Create Alert](#)

Core monitoring for Azure services

Collects metrics, activity logs, and diagnostic logs

Use for time critical alerts and notifications

Understand Azure Monitor Components



Define Metrics and Logs



- Metrics are numerical values that describe some aspect of a system at a point in time
- They are lightweight and capable of supporting near real-time scenarios

- Logs contain different kinds of data organized into records with different sets of properties for each type
- Telemetry (events, traces) and performance data can be combined for analysis

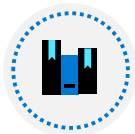
Identify Data Types



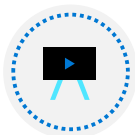
Application monitoring data – Performance and functionality of the code you have written, regardless of its platform



Guest OS monitoring – Azure, another cloud, or on-premises



Azure resource monitoring



Azure subscription monitoring – Operation and management of an Azure subscription, as well as data about the health and operation of Azure itself



Azure tenant monitoring – Operation of tenant-level Azure services, such as Azure Active Directory

Describe Activity Log Events

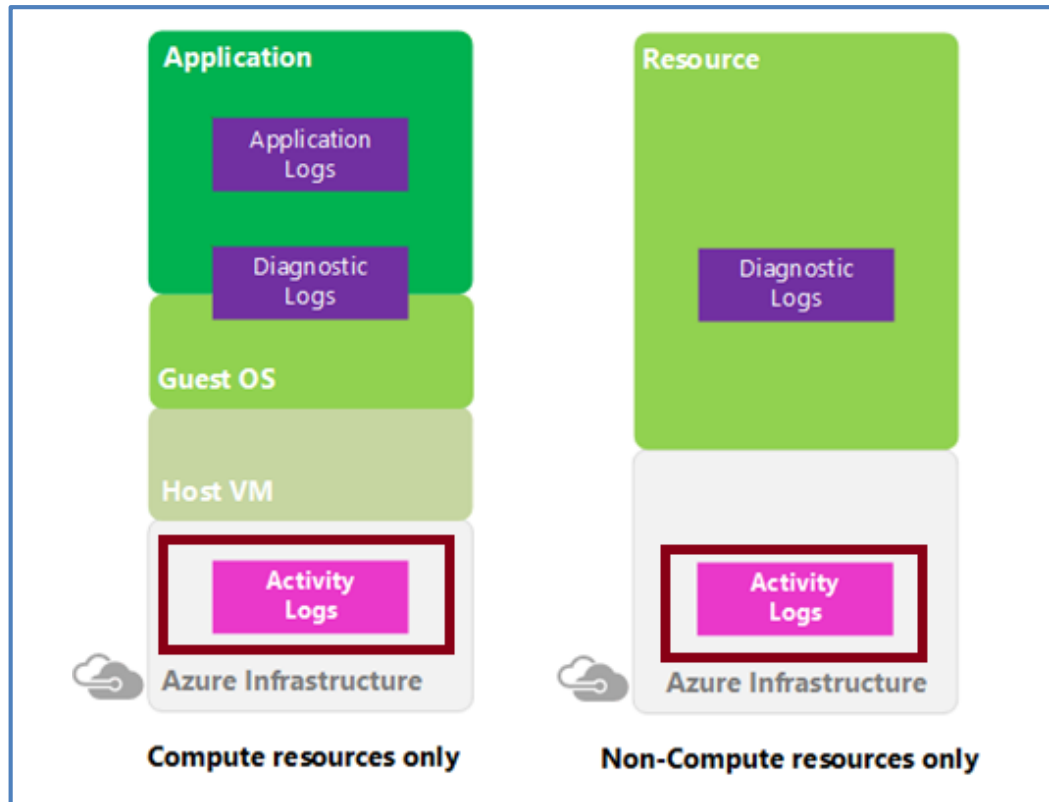
Send data to Log Analytics for advanced search and alerts

Query or manage events in the Portal, PowerShell, CLI, and REST API

Stream information to Event Hub

Archive data to a storage account

Analyze data with Power BI



Query the Activity Log

Activity log

 Edit columns  Refresh  Diagnostics settings  Download as CSV  Logs |  Pin current filters

 Search

 Quick Insights

 Add Filter

Management Group : **None**

Subscription : **2 selected**

Timespan : **Last 6 hours**

Event severity : **All**

Operation name	Status	Time	Time stamp	Subscription
>  Create or Update Virtual Network Subnet	Failed	a minute ago	Thu Mar 12 ...	ASC DEMO
>  Write GuestConfigurationAssignments	Succeeded	17 minutes ...	Thu Mar 12 ...	ASC DEMO
>  Gets workflow recommend operation groups	Succeeded	29 minutes ...	Thu Mar 12 ...	ASC DEMO

Filter by Management group,
Subscription, Timespan, and
Event Severity

Add a filter, like Event
Category (Security,
Recommendations, Alerts)

Pin current filters and
download as CSV

Lesson 02: Configure Azure Alerts



Manage Azure Monitor Alerts

Alerts

[+ New alert rule](#) [⚙️ Manage alert rules](#) [👤 Manage actions](#) [🔔 View classic alerts](#) [🔄 Refresh](#) [😊 Provide feedback](#)

Total alerts

1179

Since 2/11/2020, 11:07:58 AM

Smart groups (Preview) ⓘ

3

99.75% Reduction

Total alert rules

9

Enabled 7

Action rules (preview) ⓘ

0

Enabled 0

Severity	Total Alerts	New	Acknowledged	Closed
Sev 0	0	0	0	0
Sev 1	0	0	0	0
Sev 2	0	0	0	0
Sev 3	1178	1178	0	0
Sev 4	1	1	0	0

Unified authoring
experience

Displayed by severity

Categorized by New,
Acknowledged, and Closed

Create Alert Rules

Scope: Target selection, Alert criteria, and Alert logic

Alert rule details: name, description, and severity (0 to 4)

Action group: Notify your team via email and text messages or automate actions using webhooks and runbooks

[Home](#) > [Alerts](#) >

Create alert rule

Rules management

Create an alert rule to identify and address issues when important conditions are found in your monitoring data. When defining the alert rule, check that your inputs do not contain any sensitive content.

Scope

Select the target resource you wish to monitor.

Resource

No resource selected yet

[Select resource](#)

Condition

Configure when the alert rule should trigger by selecting a signal and defining its logic.

Condition name

No condition selected yet

Action group

Send notifications or invoke actions when the alert rule triggers, by selecting or creating a new action group

Action group name

No action group selected yet

Create Action Groups

Configure the method in which users will be notified when the action group triggers

Configure the method in which actions are performed when the action group triggers

Notifications

Configure the method in which users will be notified when the action group triggers. Select notification types, provide receiver details and add a unique description. This step is optional.

Notification type ⓘ

Name ⓘ

Selected ⓘ

Notification type ⓘ	Name ⓘ	Selected ⓘ
Email Azure Resource Manager Role Email/SMS message/Push/Voice		

Actions

Configure the method in which actions are performed when the action group triggers. Select action types, fill out associated details, and add a unique description. This step is optional.

Action type ⓘ

Name ⓘ

Selected ⓘ

Action type ⓘ	Name ⓘ	Selected ⓘ
Automation Runbook Azure Function ITSM Logic App Secure Webhook Webhook		

Lesson 03: Configure Log Analytics

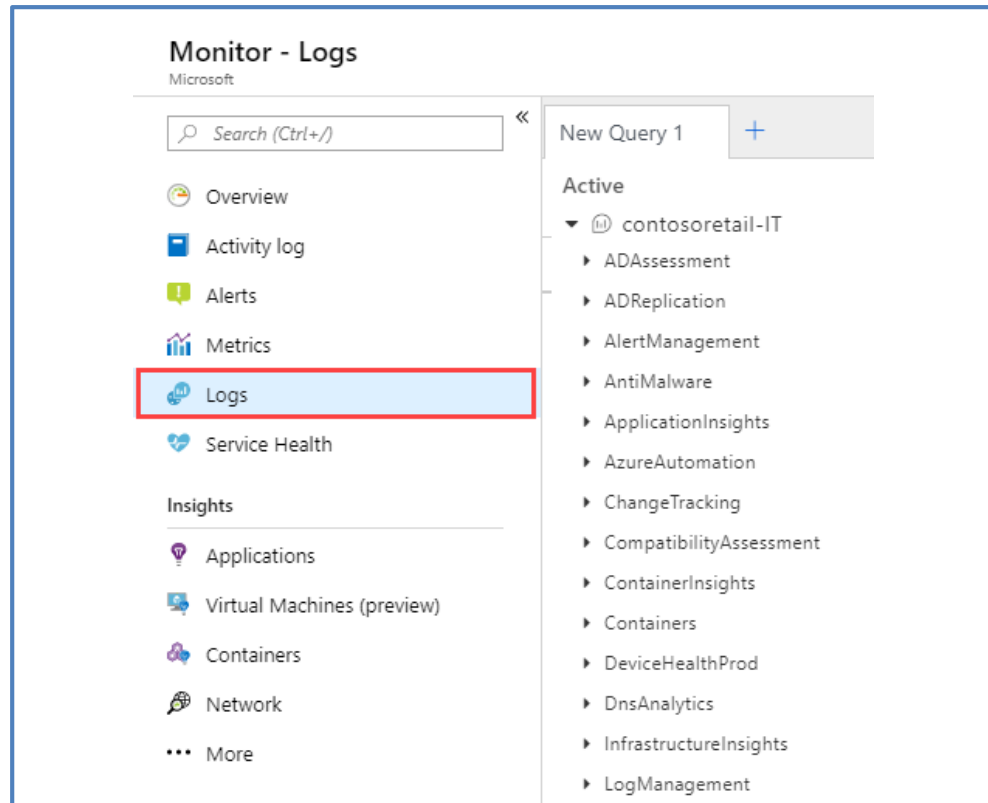


Determine Log Analytics Uses

A service that helps you collect and analyze data generated by resources in your cloud and on-premises environments

Write log queries and interactively analyze their results

Examples include assessing system updates and troubleshooting operational incidents



Create a Workspace

A workspace is an Azure resource and is a container where data is collected, aggregated, analyzed, and presented

You can have multiple workspaces per Azure subscription, and you can have access to more than one workspace

A workspace provides a geographic location, data isolation, and scope

Log Analytics workspace ☐

Create new or link existing workspace

☒ Create New ☐ Link Existing

Log Analytics Workspace * ⓘ

enter workspace name

Subscription *

Azure Pass - Sponsorship

Resource group *

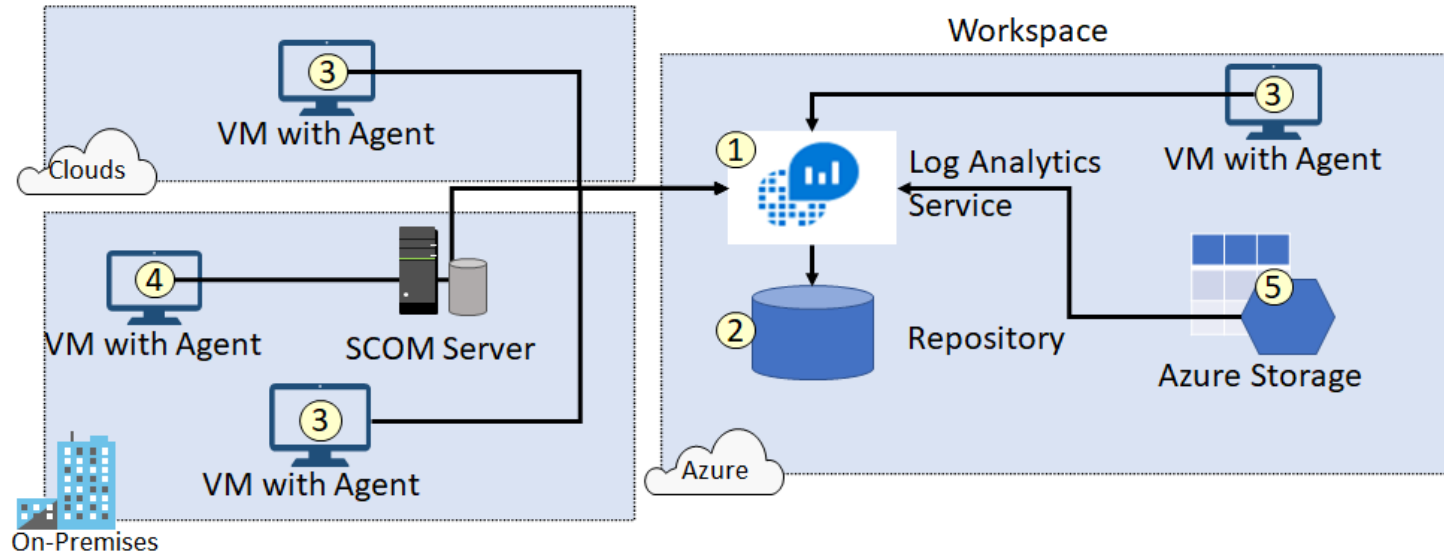
Select existing...

[Create new](#)

Location *

West US

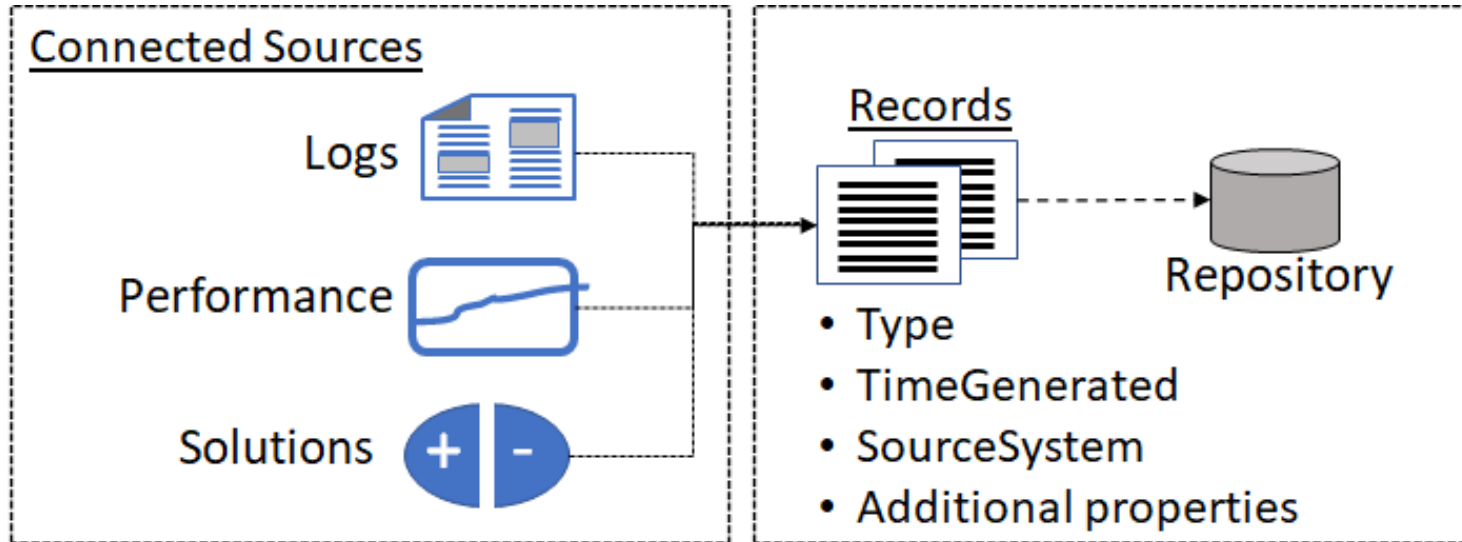
Create Connected Sources



Connected Sources generate data

Data can be collected from Windows, Linux, SCOM and Azure Storage

Define Data Sources



Data sources include Windows Event Logs, Windows Performance Counters, Linux Performance Counters, IIS Logs, Custom Fields, Custom Logs, and Syslog

Each data source has additional configuration options

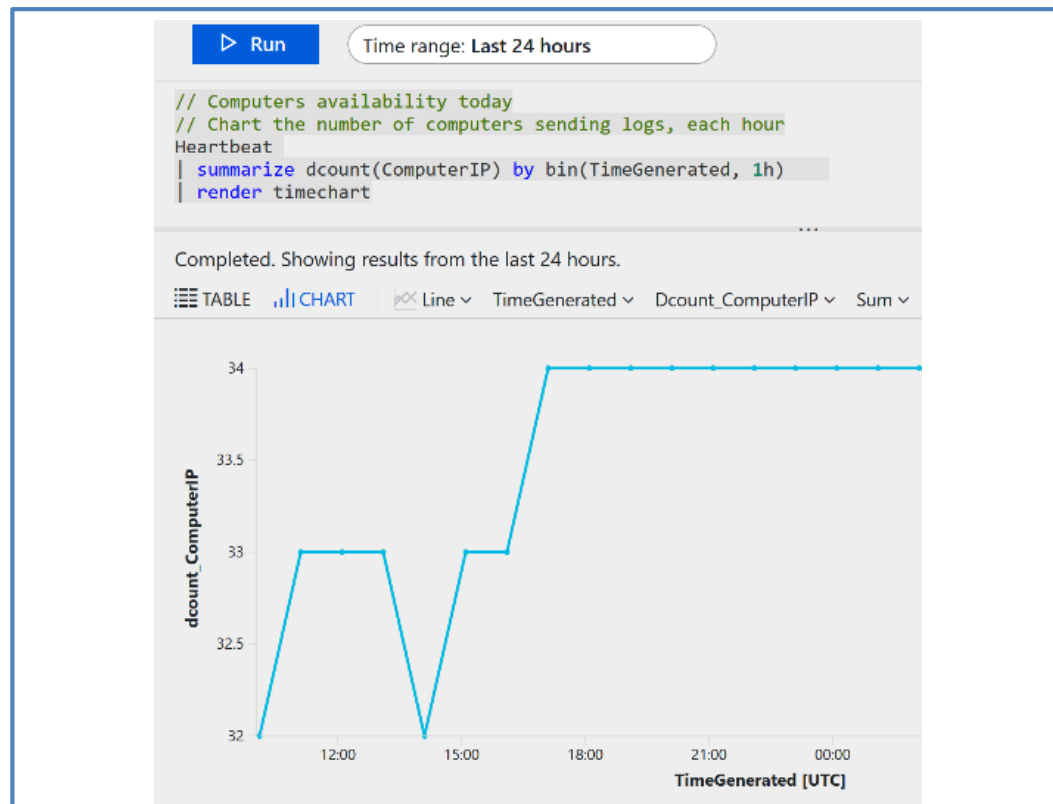
Visualize Log Analytics Data

Log Analytics provides a query syntax

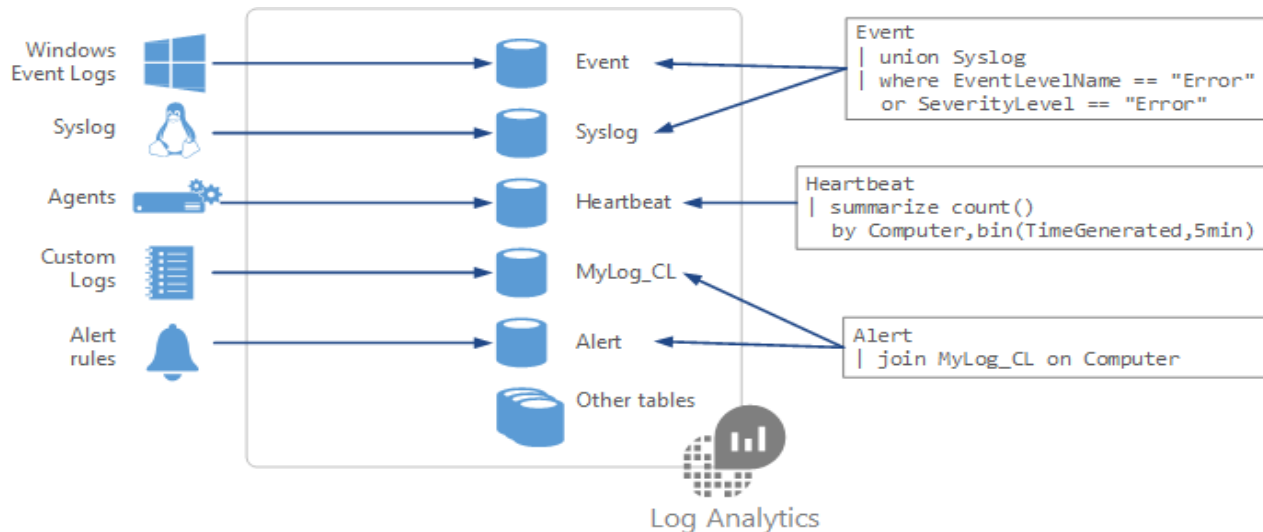
Quickly retrieve and consolidate data in the repository

Save or have log searches run automatically to create an alert

Export the data to Power BI or Excel



Structure Log Analytics Queries



```
Event  
| where (EventLevelName == "Error")  
| where (TimeGenerated > ago(1days))  
| summarize ErrorCount = count() by Computer  
| top 10 by ErrorCount desc
```

Lesson 04: Configure Network Watcher



Describe Network Watcher Features

A **regional service** that provides various network diagnostic and monitoring tools

IP Flow Verify diagnoses connectivity issues

Next Hop determines if traffic is being correctly routed

VPN Diagnostics troubleshoots gateways and connections

NSG Flow Logs maps IP traffic through a network security group

Connection troubleshoot shows connectivity between source VM and destination

Topology generates a visual diagram of resources

Network Watcher

Monitoring

 Topology

 Connection monitor

 Network Performance Monitor

Logs

 NSG flow logs

 Diagnostic logs

 Traffic Analytics

Network diagnostic tools

 IP flow verify

 Next hop

 Effective security rules

 VPN troubleshoot

 Packet capture

 Connection troubleshoot

Review IP Flow Verify Diagnostics

Checks if a packet is allowed or denied to or from a virtual machine

Network diagnostic tools

- IP flow verify
- Next hop
- Effective security rules
- VPN troubleshoot
- Packet capture
- Connection troubleshoot

Metrics

- Usage + quotas

Logs

- NSG flow logs
- Diagnostic logs
- Traffic Analytics

Packet details

Protocol
☒ TCP ☐ UDP

Direction
☒ Inbound ☐ Outbound

Local IP address * ⓘ
10.1.1.4

Local port * ⓘ
3389

Remote IP address * ⓘ
13.24.35.46

Remote port * ⓘ
3389

Check

 Access denied

Security rule
DenyAllInBound

Review Next Hop Diagnostics

Helps with determining whether traffic is being directed to the intended destination by showing the next hop

Subscription * ⓘ

MSDN Platforms Subscription

Resource group * ⓘ

Demo

Virtual machine * ⓘ

vm01

Network interface *

vm01165

Source IP address * ⓘ

10.1.1.4

Destination IP address * ⓘ

13.24.35.46

Next hop

Result

Next hop type

None

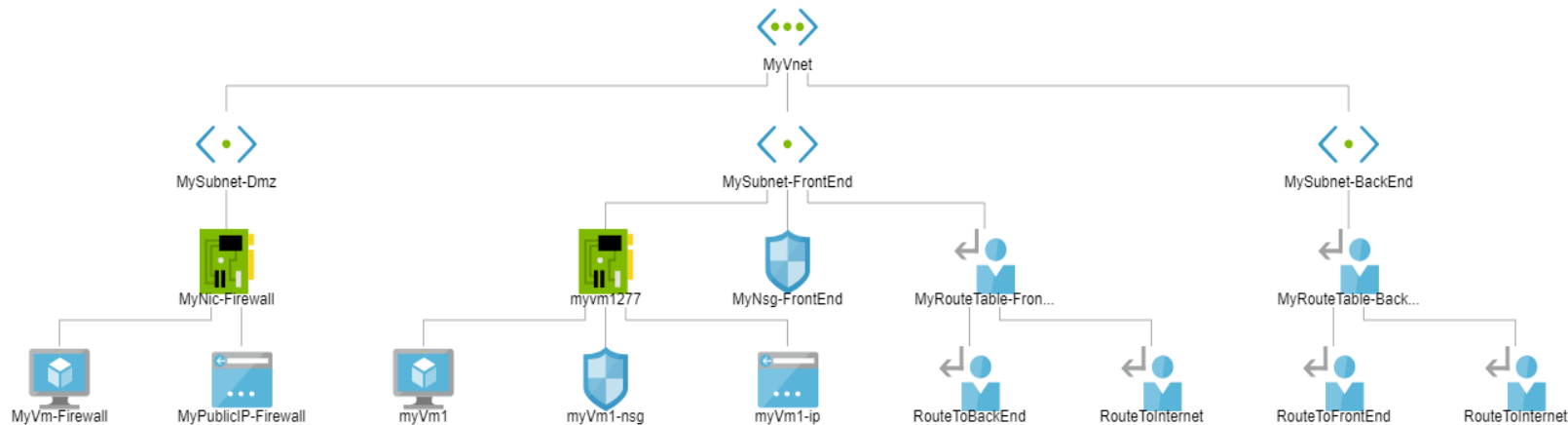
IP address

10.1.1.100

Route table ID

/subscriptions/2301e3a0-8420-... 

Visualize the Network Topology



Provides a visual representation of your networking elements

View all the resources in a virtual network, resource to resource associations, and relationships between the resources

The Network Watcher instance in the same region as the virtual network

Thank you

